

IIT Madras

ONLINE DEGREE

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PCA in higher dimensions

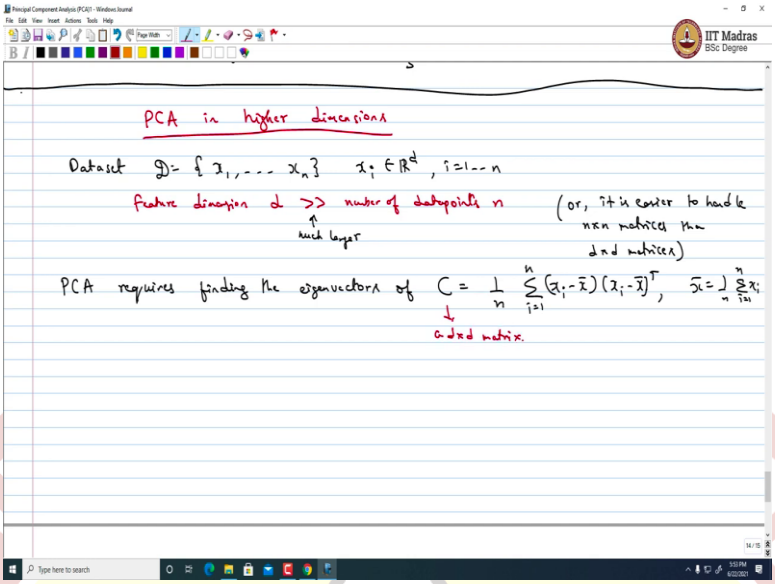
So, the last hop in the linear algebra segment of this course. This is lecture 6 and week 6, we will be looking at one final thing about principal component analysis. A discussion that we started three lectures ago. In today's lecture, we will understand how PCA can be done efficiently in higher dimensions, what I mean is given data set with data points say in some d dimensional space, where d is assumed to be large, and suppose you are given n points.

So, consider the situation where the features are high dimensional, this is input points are features. And, but the number of points are small. So, it is a lot efficient to work with, say n × n matrices as compared to d ×d while doing all computations in particular computing the eigenvalues and eigenvectors.

Now, if I were to apply the PCA algorithm naively as it is, then I will be working on d × d matrices and that to take up a lot of computation. In this lecture, we will illustrate a very neat trick where instead of working with d × d matrices, I can work with some n × n matrix, I can compute its eigenvalues, eigenvectors and still be able to achieve what PCA would do for a d × d matrix.

So, this is the so-called trick where you can save computation by working with smaller dimensional matrices where the dimension is gone by the number of data points in a situation where the data is coming from a high dimensional space.

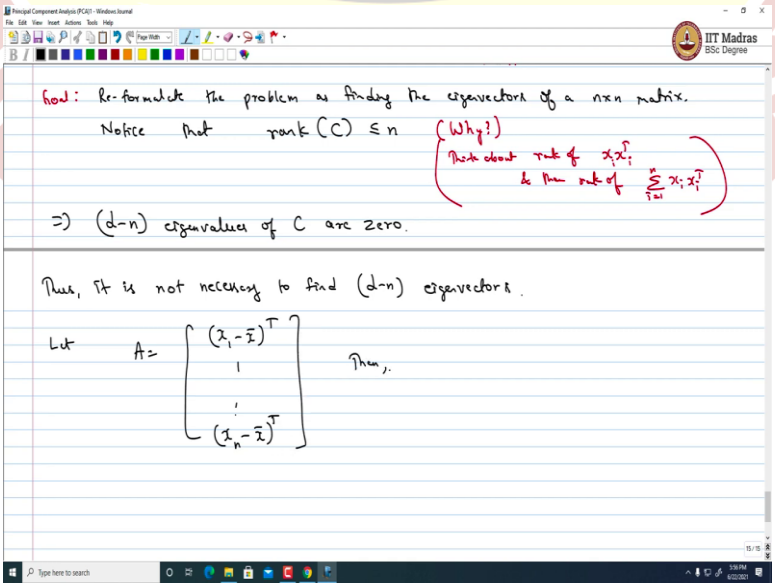
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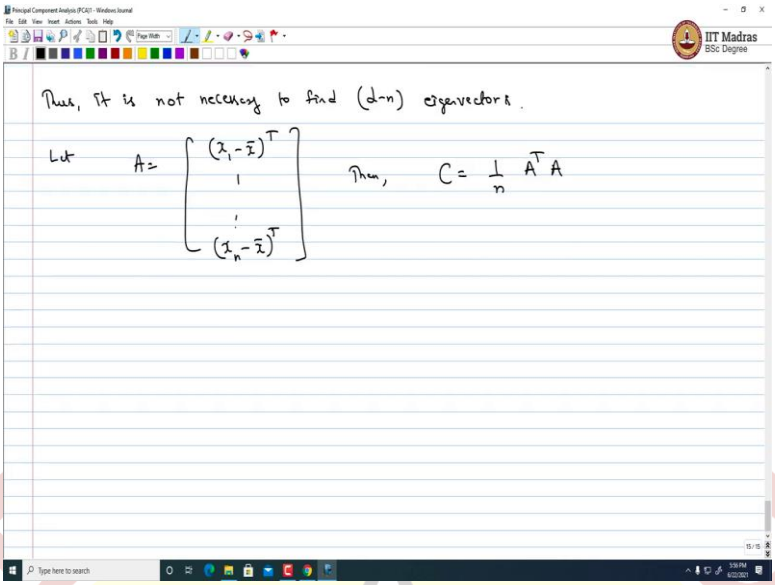


One last bit about PCA, which is to understand how to do PCA in higher dimensions, and here is where and see the connection to SVD. So, recall the problem you have a data set $\{x_1 \ldots x_n\}$, $x_i \epsilon$ to $\mathbb{R}^d$,(i = 1….. n). Now, imagine a situation where the feature dimension d is much larger than number of data points n, so this is much larger. So or it is easier to handle n ×n matrices than d × d matrices, so d × d is looking at which matrices.

So, in this case, now, if you recall PCA, PCA what does it require, requires finding the eigenvectors of the matrix C which is $\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^T$. So, $\bar{x}$ is the average. Here, $\bar{x}$ if you recall is $\sum_{i=1}^{n}(x_i)$. Now, this is a d × d matrix.

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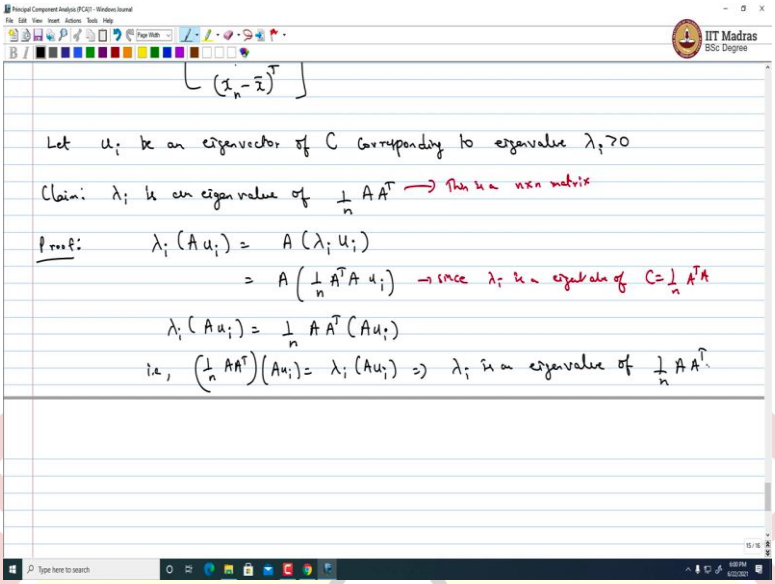


So, now what can we do now? The goal now is to reformulate the problem as finding the eigenvectors of a n × n matrix. If we can do this, if we can implement PCA by finding eigenvectors of a n × n matrix then it is going to be a lot more efficient than the original PCA algorithm which has to find the eigenvalues and eigenvectors of the d × d matrix C.

Now, notice that the rank of this matrix C is less than or equal to n. Now, this is something that you wanted to think about by just looking at this points you are adding what is the rank of $x_i x^T$, let us say something $x_i x_i^T$ and then rank of $\sum_{i=1}^{n} x_I x_i^T$.

Now, any vector with some other vector transpose even $(uv)^T$, if you look at it is a rank-1 matrix, and then if you add n such rank 1 matrices, then the rank of the sum is going to be at most n. Now, what does this imply, this implies (d – n) eigenvalues of C are 0, as simple as that. Thus, it is not necessary to find (d – n) eigenvectors. So, let we will write this matrix A as $\begin{bmatrix} (x_1 - \bar{x})^T \\ \vdots \\ (x_n - \bar{x})^T \end{bmatrix}$. Then, most important C can be written as $\frac{1}{n}.A^T A$.

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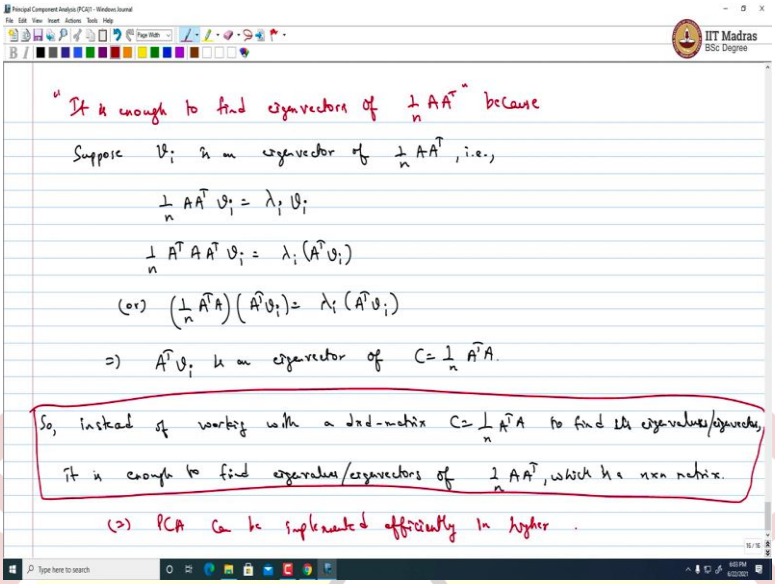


Now, we still have not solved any dimension problem, but we have just rewritten the same problem. Now, let u_i be an eigenvector of C corresponding to eigenvalue $\lambda_i \geq 0$. Now, we claim λ_i is an eigenvalue of $\frac{1}{n}$A A^T. And the reason why we are doing the switch is because this is a n × n matrix and A A^T. Whereas, A^T A is a d × d matrix.

Now, the proof is short and simple you have λ_i (Au_i) is the same as just A(λ_i u_i) Now, λ_i is a eigenvalue of A^T A. So, I have A ($\frac{1}{n}$ A^T A u_i). Since λ_i is an eigenvalue of C=($\frac{1}{n}$ A^T A.) Now λ_i (Au_i) is nothing but $\frac{1}{n}$A A^T(A u_i).

Now, if you read this might your equation the other way that is $\frac{1}{n}$ A A^T applied on some vector $Au_i = \lambda_i$ (Au_i) implies λ_i is an eigenvalue of $\frac{1}{n}$ A A^T .

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Now, next important claims it is enough to find eigenvectors of $\frac{1}{n}$ A^T A. Because now if you imagine you have got an eigenvector. Let us say, so the eigenvalues are the same. Now, I just need to find the eigenvectors. Suppose, v_i is an eigenvector of $\frac{1}{n}$ A A^T, that is

$\frac{1}{n}$ A A^T v_i = λ_i v_i. What I am going to do is $\frac{1}{n}$ $A^T A A^T v_i$ = λ_i (A^T v_i) or ($\frac{1}{n}A^T A$) ($A^T v_i$)is nothing but λ_i (A^T v_i), implies $A^T v_i$ is an eigenvector of C = $\frac{1}{n}$ A^T A. So, that is about it. So, instead of working with a d × d matrix C = $\frac{1}{n}$ A^T A, to find its eigenvalues slash eigenvectors it is enough to find eigenvalues slash eigenvectors of $\frac{1}{n}$ A A^T which is a n × n matrix.

Now, this is the punch line. So, PCA can be implemented efficiently in higher dimensions by just working with, point 2, PCA can be implemented efficiently in higher dimensions. That is the point of this passage. A recap before we end the discussion about PCA. So, what have we learned in today's lecture? We learned that one could apply PCA or implement PCA in higher dimensions efficiently.

Efficiency in terms of computation by I mean for a given data set with d-dimensional points and totally n points given, one could work with n × n matrices, which is much lower dimension, lower-dimensional quantities as compared to d × d. Compare its eigenvalues, eigenvectors and still be able to achieve exactly what PCA does with a d × d matrix.

So, at no point, we are working with a d × d matrix to compare its eigenvalues or eigenvectors instead I just work with n × n matrix and solve this problem efficiently in higher dimensions. So, that is the main message of this lecture.